

Total No. of Questions : 10]

SEAT No. :

P3174

[Total No. of Pages : 2

[5354]-691

B.E. (Information Technology)
INFORMATION AND CYBER SECURITY
(2012 Pattern) (Semester - I)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8, Q9 or Q10.
- 2) Figures to the right indicate full marks.

- Q1)** a) Distinguish between Symmetric and Asymmetric key cryptography. Explain the principle of one time pad with example. [6]
b) In RSA, if public key $e = 7$ and modulus $n = 33$, calculate private key d . [4]

OR

- Q2)** a) Draw AES block diagram and state the general steps in detail. [6]
b) Using Euclidean algorithm calculate following [4]
i) GCD (88, 220)
ii) GCD (25, 60)

- Q3)** a) Explain X.509 standard for digital certificate. [6]
b) Find the value of x using Chinese remainder theorem; [4]
 $x \equiv 2 \pmod{3}$, $x \equiv 3 \pmod{5}$, $x \equiv 2 \pmod{7}$

OR

- Q4)** a) Explain the difference between MAC and Message Digest with example. [6]
b) Compare and contrast MD5 and SHA1. [4]
- Q5)** a) How AH and ESP are differs while working under transport and tunnel mode. [8]
b) Explain different IDS methods with one example each. [8]

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OR

- Q6)** a) Define IKE protocol and explain its aggressive mode and quick mode in brief. [8]
b) Explain SSL hand shake and SSL Record protocol in detail with neat sketch. [8]
- Q7)** a) Describe the classification of cybercrime. [10]
b) Describe the Indian and global legal perspective on cyber-crime [6]

OR

- Q8)** a) What is cyber stalking? Explain the types of stalkers. [10]
b) Address security issues in cloud computing. [6]
- Q9)** a) Describe following in detail [12]
i) Indian IT ACT 2000 and its challenges.
ii) SQL injection.
b) Distinguish between phishing and pharming? Describe key loggers and Spywares in brief. [6]

OR

- Q10)** Write short note on following : [18]
a) Different ways of password cracking
b) Host based malicious programs: Trap Door, Logic Bombs and Trojan horse
c) Proxy server and an anonymizer

